SE(3)-WAM: Forecasting the World at Action Time for Dynamic Manipulation

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Historical research draft — results pending.
Method and experiment descriptions are candidates, not validated findings.

***Abstract*— Actions in dynamic manipulation take effect after sensing, inference, queuing, and actuation delays, when the observed scene is already stale. This draft proposes SE(3)-WAM, a feed-forward world-action model that forecasts the physical world at the times its commands will execute. From timestamped two-view RGB-D, proprioceptive, and action history, SE(3)-WAM predicts target-centric multi-horizon $\mathrm{SE}(3)$, dense metric 3-D flow, and calibrated uncertainty. It compresses these forecasts into a fixed set of physical tokens that condition a flow-matching action expert through a one-way, directly intervenable bridge; a newest-wins executor then applies only temporally valid actions. We further introduce DB0, a replayable dynamic-manipulation benchmark with exact-current counterfactual histories, controlled latency and time-to-contact, orientation and occlusion contrasts, free dynamics, and contact-triggered replanning. Planned comparisons test whether explicit futures improve on matched history-aware policies and whether swap, zero, noise, and oracle interventions identify useful causal use. The result tables remain pending; no success-rate gain, mechanism benefit, or transfer to external benchmarks or a real conveyor is established by this draft.**

I. Introduction

A robot never acts on the instant it observes. Camera exposure, transfer, inference, command queuing, and actuation introduce a capture-to-act delay during which a target can translate, rotate, become occluded, or change motion after contact. Faster inference and continuous action streaming reduce this delay [1], [2], but cannot remove it or recover motion that is hidden from the latest frame. Dynamic manipulation therefore requires a policy to answer a physical-time question: *where will the relevant geometry be when this action is applied?*

Recent systems address important parts of this problem. DynamicVLA and VLASH reduce observation age through asynchronous chunk execution [1], [2]. PUMA uses future-feature prediction as training supervision [3]; DBC-TFP and AHEAD instead feed a predicted image or latent future into the policy [4], [5]. PhysMani and LaMP bring 3-D motion into action generation [6], [7]. The missing capability is a single feed-forward system that infers motion from raw timestamped RGB-D, predicts independently measurable metric geometry at measured execution times, represents uncertainty, and exposes a clean intervention point between world forecasting and action.

We introduce **SE(3)-WAM**, an execution-aligned world-action model for this setting (Fig. 1). A temporal metric 3-D encoder lifts frozen visual features with measured depth and camera geometry. Its world expert predicts future target transforms on $\mathrm{SE}(3)$, dense scene displacement, and heteroscedastic uncertainty at eight physical-time horizons. A target- and mover-aware reducer turns these outputs into one compact token per execution horizon. The action expert retains the current scene representation and reads the physical future through a one-way bridge, making the forecast simultaneously useful for control, measurable in physical units, and directly intervenable. During continuous execution, stale action prefixes are discarded and overlapping commands are replaced by the newest temporally valid chunk.

The paper makes three contributions:

- **Action-time world forecasting:** a feed-forward formulation that predicts target-centric multi-horizon $\mathrm{SE}(3)$, dense metric 3-D flow, and calibrated uncertainty from non-privileged timestamped RGB-D history at the times commands will execute;
- **An actionable physical interface:** a fixed-size, one-way bridge from forecast to action whose zero, swap, noise, and oracle interventions distinguish causal use of foresight from auxiliary representation learning; and
- **DB0:** a causal benchmark with fixed physical-time conditions, minimal observation–action trajectories, exact-current counterfactuals, dynamic manipulation, external protocols, and separate real-conveyor evaluation.

The planned comparisons ask whether this candidate improves closed-loop manipulation over matched history-aware streaming policies and whether forecast interventions change behavior in a useful direction. These are hypotheses; the pending manuscript results do not yet distinguish a stronger temporal representation from causal use of physically correct foresight.

II. Related Work

A. Dynamic Manipulation and Real-Time Execution

Classical dynamic grasping couples pose prediction with reachability-aware planning [8]; visual servoing makes motion locally quasi-static, as GEM scales to conveyors [9]. Learned systems reduce observation age: DynamicVLA lets the world evolve during inference and drops stale actions [1], VLASH rolls state forward [2], and AHA-WAM splits slow planning from fast action [10]. PUMA/DOMINO and PhysMani report latency [3], [6], but their public protocols do not define DB0's comparable per-episode clock ledger or controlled realized-delay factor.

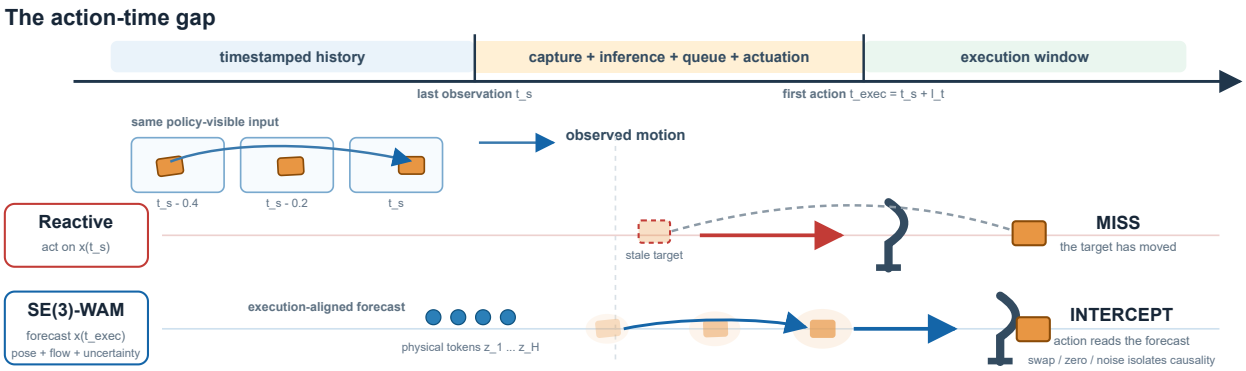


Fig. 1. Why action-time forecasting matters. Both policies observe the same timestamped history. A reactive target becomes stale during the measured capture-to-act delay, whereas SE(3)-WAM forecasts target pose, scene motion, and uncertainty at the execution window and conditions the action on that future.

B. Future Supervision and Forecast-to-Action Models

World-action models often use future prediction to shape representations without executing the predicted future. PUMA predicts pooled future DINO features from historical optical flow and world queries [3]; GaussianDream and EgoWAM study 3-D or geometric future supervision that can be removed at inference [11], [12]. Fast-WAM further shows that co-training can outperform test-time imagination in some regimes [13]. Our matched *joint auxiliary* control separates this representation benefit from the benefit of exposing a predicted future to the action policy.

Other methods make foresight an inference-time input. DBC-TFP fuses a predicted future RGB image with the current image [4]. AHEAD predicts selected future OpenVLA patch tokens using 2-D flow-derived kinematics and uncertainty-aware horizon stopping [5]. LaWAM conditions action on latent future subgoals [14], while FutureRTC anticipates the execution-time latent state [15]. Making Foresight Actionable and OA-WAM show that an explicit, intervenable world–action interface is needed to demonstrate causal use of a forecast [16], [17]. SE(3)-WAM realizes this principle with metric 3-D quantities tied to measured execution times.

ACT, Diffusion Policy, and DP3 establish strong sequence and metric-3-D action models [18]–[20]. The proposed comparison holds the action objective, chunk, and 3-D input fixed to distinguish a possible contribution of execution-aligned physical futures from a stronger action decoder.

C. Metric 3-D World-Action Modeling

PhysMani forecasts a divergence-free velocity field over canonical 3-D Gaussians and conditions a policy through local cross-attention [6]. LaMP uses 3-D scene-flow hidden states as a latent motion prior [7]; PointWorld predicts action-conditioned dense 3-D point flow for model-predictive control [21]; object-centric Gaussian models forecast multiple rigid-body transforms under candidate actions [22]. 3DFlowAction predicts desired object flow as an action representation rather than exogenous evolution [23]. SE(3)-WAM unifies raw timestamped RGB-D history, target-centric metric $\mathrm{SE}(3)$, dense flow, measured execution horizons, uncertainty, and a compact intervenable bridge. Table I shows how this combination connects physical forecasting, real-time execution, and causally auditable action.

III. SE(3)-WAM

A. Physical-Time Problem Formulation

At policy query t, the SE(3)-WAM loader assembles a masked, timestamped history $\mathcal{H}_t$ from two-view RGB-D packets, camera metadata, proprioception, and past high-level actions. Resulting motion is observed through subsequent robot state. These are method-side buffering choices, not DB0 policy-packet fields. The most recent sensor timestamp is τ_t^s. In DB0, the capture-to-act delay ℓ_t is the registered episode condition; the real-system worker measures its variable counterpart. The h-th world target is evaluated at

$$\tau_h^\star = \tau_t^s + \ell_t + h\Delta_a, \qquad h = 0, \dots, H-1, \tag{1}$$

where Δ_a is the controller period. Thus a fixed model index denotes a physical execution time, not merely "h video frames ahead." The model predicts the task-object transform $\hat{\boldsymbol{T}}_h \in \mathrm{SE}(3)$, patch displacement $\hat{\boldsymbol{f}}_{i,h} \in \mathbb{R}^3$, uncertainty $\hat{\boldsymbol{\sigma}}_h$, and an action chunk $\hat{\boldsymbol{a}}_{0:H_a-1}$. Future labels after robot contact are masked unless they are generated under the same stored high-level actions, controller, and fixed-delay condition; this prevents an unconditional model from being scored against a world altered by its own action.

a) Clock and label construction: DB0 aligns training labels from capture time, the episode's fixed delay, and the physics trace. Translation is linear in the base frame, whereas rotation uses $\boldsymbol{R}_j \operatorname{Exp}(\alpha \operatorname{Log}(\boldsymbol{R}_j^\top \boldsymbol{R}_{j+1}))$; object symmetry is applied only by the loss, never by altering the label. Simulated flow transports visible depth samples through persistent body identity into the same base frame. A point is invalidated when its correspondence is undefined, and an entire future is invalidated once unmatched robot contact makes it action-dependent. Training samples delays from the registered training support and supplies ℓ_t as a continuous

TABLE I

WORLD-ACTION MODEL CAPABILITIES. "CONSUMED" DENOTES AN ABLATABLE INFERENCE PATH FROM PREDICTED FUTURE TO ACTION; METRIC 3-D DENOTES FORECASTS THAT ARE INDEPENDENTLY MEASURABLE IN PHYSICAL UNITS.

Method	History	Metric 3-D	Target $\mathrm{SE}(3)$	Future consumed	Uncertainty	Feed-forward	Principal boundary
DBC-TFP [4]	✓	–	–	✓	–	✓	Future RGB; controlled simulation
PUMA [3]	✓	–	–	–	–	✓	Future-feature auxiliary loss
DynamicVLA [1]	✓	–	–	–	–	✓	Implicit history-to-action streaming
AHEAD [5]	✓	–	–	✓	✓	✓	2-D latent kinematic extrapolation
PhysMani [6]	✓	✓	–	✓	–	–	Per-scene reconstruction/optimization
LaMP [7]	✓	✓	–	✓	–	✓	Latent scene-flow prior
SE(3)-WAM	✓	✓	✓	✓	✓	✓	Action-time metric forecast with a causal bridge

input. Variable capture/availability/issue/application timing is logged only by the separate asynchronous real-system worker; it is not reconstructed or duplicated in the core DB0 trajectory.

B. Timestamped Metric 3-D Encoder

For each view and valid history time, a frozen DINOv3 encoder [24] produces a 14×14 patch grid. Measured depth and camera intrinsics lift patch centers into 3-D; time-varying extrinsics map them to the robot base frame. A token combines appearance, base-frame position, view ID, true time offset $\tau_k^s - \tau_t^s$, frame age, and a validity bit. Separate tokens encode robot state, gripper state, and the stored high-level action history. Target-query structure is fixed by the training/evaluation configuration and receives no task identifier, natural-language description, or future state.

A factorized transformer first performs within-frame spatial fusion and then temporal attention at matched patch/view locations. Unlike repeat padding, an explicit history-valid mask preserves the information boundary at early task steps. Base-frame 3-D positional encoding makes metric depth and camera motion available without treating depth as a third RGB channel. The encoder outputs scene tokens $\boldsymbol{X}$ and H target queries $\boldsymbol{q}_{1:H}$.

a) Information boundary: DB0's standard policy packet contains only timestamped RGB-D, calibration metadata, and robot state; task identity selects the evaluator outside the policy. The SE(3)-WAM loader locally constructs history, validity masks, and action memory. Masks, object/goal/contact/future state, factors, material and motion-generator variables, RNG state, and the T5 event tape remain private supervision, teacher, or evaluator records. Prediction-only P0 samples carry no action target and therefore contribute exactly zero action gradient. Normalization and uncertainty calibration are fitted on training records only.

C. Target-Centric Physical Forecast

The object queries estimate the latest visible target pose $\hat{\boldsymbol{T}}_t$ and a relative Lie-algebra trajectory $\hat{\boldsymbol{\xi}}_h \in \mathfrak{se}(3)$:

$$\hat{\boldsymbol{T}}_h = \mathrm{Exp}(\hat{\boldsymbol{\xi}}_h^{\wedge})\hat{\boldsymbol{T}}_t, \quad \hat{\boldsymbol{\xi}}_h = [\hat{\boldsymbol{\rho}}_h^{\top}, \hat{\boldsymbol{\phi}}_h^{\top}]^{\top}. \tag{2}$$

Translation is reported in centimeters and rotation by the symmetry-aware $\mathrm{SO}(3)$ geodesic. A diagonal tangent-space log variance accompanies every horizon. In parallel, a spatial decoder predicts metric patch displacement $\hat{\boldsymbol{f}}_{i,h} = \hat{\boldsymbol{x}}_i(\tau_h^{\star}) - \boldsymbol{x}_i(\tau_t^s)$ and log variance. Pose and flow are complementary: $\mathrm{SE}(3)$ yields an action-ready rigid target state, whereas dense flow exposes non-target movers, depth motion, and consistency violations.

For a residual $\boldsymbol{e}$ with predicted log variance $\boldsymbol{s}$, we use the heteroscedastic penalty

$$\mathcal{L}_{\mathrm{het}}(\boldsymbol{e}, \boldsymbol{s}) = \tfrac{1}{2}\sum_j \left[\exp(-s_j)e_j^2 + s_j\right]. \tag{3}$$

The pose loss applies Eq. (3) to translation and to $\mathrm{Log}(\hat{\boldsymbol{R}}_h^{\top}\boldsymbol{R}_h)^{\vee}$ after minimizing over the declared object symmetry group. The flow loss applies it only to valid, exogenous points. A rigid-consistency term aligns target-region flow with the displacement induced by $\hat{\boldsymbol{T}}_h\hat{\boldsymbol{T}}_t^{-1}$.

D. Intervenable Physical Summary and Action Expert

Passing all spatiotemporal flow tokens to the policy would obscure the interface and increase latency. We instead form one summary token per execution horizon:

$$\boldsymbol{z}_h = g_{\psi}\Big(\mathrm{Log}(\boldsymbol{T}_{E,t}^{-1}\hat{\boldsymbol{T}}_h), \hat{\boldsymbol{v}}_h, \operatorname*{Pool}_i w_{i,h}\hat{\boldsymbol{f}}_{i,h}, \hat{\boldsymbol{\sigma}}_h, \tau_h^{\star}, \rho_{\mathrm{react}}\Big), \tag{4}$$

where $\boldsymbol{T}_{E,t}$ is the end-effector pose, $w_{i,h}$ is a learned target/mover weight, and ρ_{react} is total delay divided by the latest feasible contact time. The H tokens preserve horizon identity and uncertainty but have a fixed compute cost independent of image resolution.

The action expert predicts an H_a-step E7 chunk (body-frame translation, axis-angle rotation, and gripper). Let $\boldsymbol{c}_t = (\boldsymbol{X}, \boldsymbol{z}_{1:H})$ denote its visual–physical context. With Gaussian source $\boldsymbol{a}_0$ and data chunk $\boldsymbol{a}_1$, conditional flow matching [25] uses

$$\begin{aligned} \boldsymbol{a}_\tau &= (1-\tau)\boldsymbol{a}_0 + \tau\boldsymbol{a}_1, \\ \mathcal{L}_{\mathrm{act}} &= \|\boldsymbol{v}_\theta(\boldsymbol{a}_\tau, \tau \mid \boldsymbol{c}_t) - (\boldsymbol{a}_1 - \boldsymbol{a}_0)\|_2^2. \end{aligned} \tag{5}$$

A one-way cross-attention adapter lets action queries read $\boldsymbol{z}_{1:H}$; scene and world tokens never read action tokens. Forecast pretraining is followed by the default detached-bridge action stage, which prevents the action objective from destroying metric semantics. A one-way end-to-end stage is evaluated separately. This design gives a precise

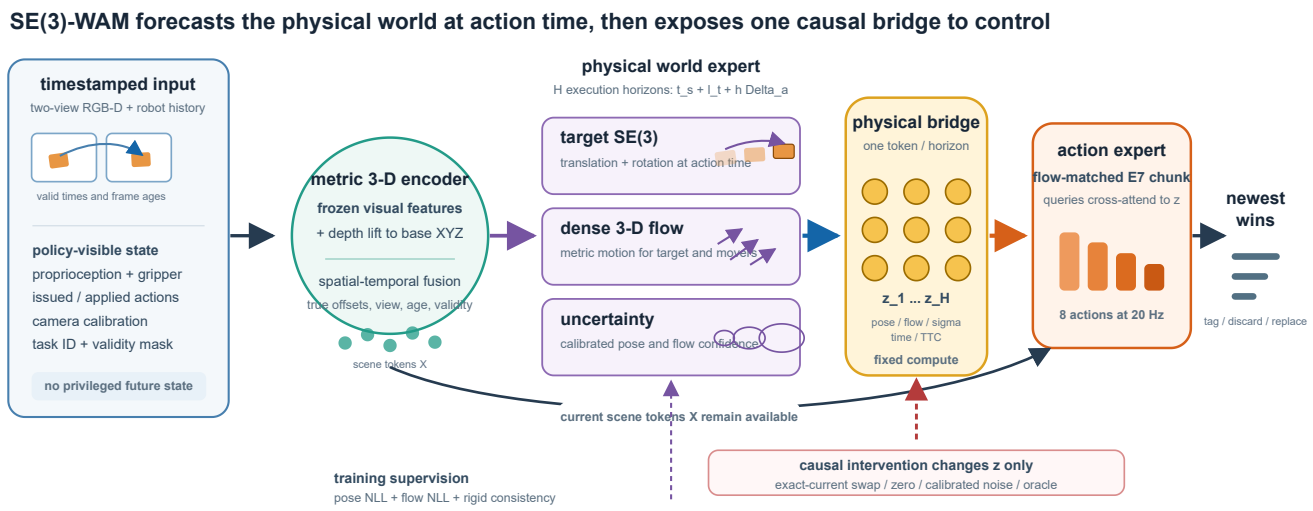


Fig. 2. SE(3)-WAM maps timestamped RGB-D history to action-time physical forecasts and continuously executed action chunks. Depth-lifted features produce target SE(3), dense 3-D flow, and uncertainty at each execution horizon; a fixed-size one-way bridge makes this future available to the action expert and provides a single causal intervention point.

intervention point: replace $\boldsymbol{z}$ by zero, a matched-sample swap, calibrated noise, or oracle future while leaving current perception unchanged.

a) Matched information and gradient paths: The mechanism comparison is defined by three conditionals with a shared encoder, world targets, action queries, and parameter envelope:

$$\begin{aligned} \text{J-Joint:} \quad & \boldsymbol{a} \sim p_\theta(\boldsymbol{a} \mid \boldsymbol{X}), && \boldsymbol{z} \sim q_\psi(\boldsymbol{z} \mid \boldsymbol{X}), \\ \text{P-Detach:} \quad & \boldsymbol{a} \sim p_\theta(\boldsymbol{a} \mid \boldsymbol{X}, \mathrm{sg}[\boldsymbol{z}]), \\ \text{P-E2E:} \quad & \boldsymbol{a} \sim p_\theta(\boldsymbol{a} \mid \boldsymbol{X}, \boldsymbol{z}), \end{aligned} \tag{6}$$

where sg stops action gradients but not forecast gradients. J-Joint measures representation shaping, P-Detach isolates inference-time consumption while preserving metric semantics, and P-E2E measures the effect of task gradients on the physical interface. The action expert retains $\boldsymbol{X}$ in all three models, and interventions change only $\boldsymbol{z}$; current perception, robot state, and action source noise therefore remain fixed.

The complete objective is

$$\mathcal{L} = \mathcal{L}_{\text{act}} + \lambda_T \mathcal{L}_{\text{pose}} + \lambda_F \mathcal{L}_{\text{flow}} + \lambda_C \mathcal{L}_{\text{cons}}. \tag{7}$$

Direct world regression is the proposed main decoder, not a measured winner. Standard world flow matching and aligned-previous-chunk Residual-SPDF are candidate controls under the same data, parameters, physical horizons, and number of function evaluations; their forecast–control–latency trade-off remains to be measured.

E. Training and Deployment Protocol

Training is staged to keep mechanism comparisons identifiable. First, the metric encoder and world expert are trained on P0 plus valid exogenous pre-contact windows from manipulation episodes. Each loss is divided by its number of valid targets, so a long visible sequence cannot outweigh an occluded or early-history example. Minibatches are balanced over task, motion level, object family, visibility, and horizon availability rather than sampled in proportion to raw episode count. Geometric augmentation is shared by both RGB-D views and their calibration; color augmentation never changes depth or supervision. Frozen DINOv3 features may be cached, but the cache key includes image, crop, encoder, and calibration versions.

Second, the action-only current and history baselines select direct regression or flow matching on the same validation manifest. This freezes the action objective, chunk, source-noise convention, and optimizer before any explicit future is consumed. J-Joint then adds the world losses without an action edge. P-Detach initializes from that checkpoint, freezes the selected physical semantics, and trains only the reducer, bridge, and action-side parameters; P-E2E is a separately seeded continuation with all metric losses still active. No variant receives extra demonstrations, task-specific checkpoints, privileged masks, or a longer history.

World-decoder families are compared on prediction, systems-profile latency, and the same closed-loop development cells under equal data, parameters, NFE, precision, and execution horizons. The final checkpoint rule, seeds, and Core manifest are fixed before Test-OOD, external, or real-robot evaluation. At deployment, only policy-visible inputs are loaded: simulator instance masks, target pose, future labels, and teacher feasibility scores are absent.

F. Continuous Execution

Inference runs in a latest-observation worker rather than waiting for each chunk to finish. Every output is tagged by observation and target control step. A completed chunk discards actions whose target time has passed and replaces older actions on overlapping future steps. The runtime-only executor logs capture, availability, inference, issue, and apply timestamps; Eq. (1) is updated from the realized delay. These

system logs are not fields in the DB0 training dataset. This combines explicit environmental forecasting with the strongest practical lesson of streaming policies: the world never pauses for the model.

IV. DB0: Dynamic Manipulation Protocol

DB0 is a compact causal benchmark for dynamic manipulation. It isolates observability, execution delay, metric future utility, orientation, occlusion, and contact under one reproducible MuJoCo 3.11/robosuite physical clock [26], [27]. Every task uses a single Franka arm, synchronized fixed and wrist RGB-D cameras at 20 Hz, 500 Hz physics, and an E7 operational-space controller. Each policy packet contains only the two RGB-D views and robot/gripper state; timestamps and camera calibration are sensor metadata. Masks and all simulator truth—object, goal, contact, future state, factors, event tapes, material, and RNG—remain privileged labels for supervision, teachers, or evaluators. DB0 streams current observations; each method buffers history and selects its context and prediction/action horizons.

Tasks. P0 evaluates physical forecasts at declared horizons without robot motion. T1-Belt intercepts a conveyor object; T1-XYZ/SO3 share centroid motion but only the latter rotates an asymmetric target, and T1-OCC adds physical occlusion. T2-TRANS/SE3 release into translating or rotating receptacles; T3-Phase/Full traverse a rotating buzz wire. T4-Sphere/Slider/Can capture rolling, sliding, or roll–slip objects leaving a ramp. After one of two striker impacts, T5-Replan must grasp the resulting keyed puck. Ordered evaluators reject touch-only, blocking, unilateral contact, freezing, and early rigid attachment. A privileged planner/IK/MPC teacher emits action demos; only the evaluator records physical events and outcomes.

Counterfactuals and splits. Exact-current groups share current pose, appearance, robot state, and task parameters while histories imply different future motion. Time-reversed and same-current swapped histories expose single-frame shortcuts. Test-ID supports static, constant-twist, and smoothly changing motion. Test-OOD changes exactly one predeclared object, spatial, or motion-physics factor; it is reported diagnostically and never used for model selection. Unannounced post-query events are evaluated as adaptation, not as deterministic forecast error.

Replay and outcome contract. Each trajectory stores synchronized two-view RGB-D, robot state, and one high-level action; frame index and capture time align steps. Episode metadata holds rates, controller, fixed delay, calibration, and versions. No issued/applied duplicate, per-step issue/apply timestamp, or 500 Hz torque trace is exported, although the simulator retains its controller and T5 queue. Object state, teacher phase, events/outcomes, and future labels are privileged. Episodes freeze base state, factors, source group, and paired identity; descendants remain in one split and all methods share the manifest. DB0 provides no context or reset pre-roll; asynchronous timing is a separate real-system profile.

Success is an ordered physical event trace rather than a final pose threshold. The common prehensile contract requires opposing finger-pad contact for at least 50 ms, loss of support and 5 cm center-of-mass clearance, then a 0.5 s stable hold while the driver keeps moving. T2 analogously requires entrance-plane crossing, full collision-geometry containment, release, low relative motion, and dwell; T3 fails on calibrated wire contact. These traces localize whether a rollout failed in forecast, feasibility, approach, first contact, grasp, transport/release, or post-event adaptation, and prevent touch, pin, stop, or fixture block from being counted as success.

Metrics and statistics. Core layers are (i) SR, hierarchical TC, success-only CT, and safety-failure rate; (ii) per-horizon translation and symmetry-aware rotation error; and (iii) SE(3)-WAM-specific EPE3D and Flow–$\mathrm{SE}(3)$ consistency. Fixed latency is an episode condition; wall-clock latency is separate systems profiling. Headline results use five seeds, paired 100-rollout cells, Wilson intervals, and clustered paired bootstrap over counterfactual groups.

External validity. We complement the controlled DB0 comparison with native protocols: a single-arm DOMINO subset [3], DOM-Dyn4 with DynamicVLA [1], all normal/high PhysMani families [6], static LIBERO compatibility [28], and 180 real Franka conveyor trials. Native scores remain separate from DB0, and every compared method is rerun under the corresponding protocol.

V. Experiments

This section preserves a proposed evaluation design. Its tables contain pending fields, not completed runs; budgets and comparisons must be reconciled with the current Research experiments before execution. We ask five questions: **Q1**, how much do timestamped history and metric 3-D improve dynamic action? **Q2**, what does an explicit predicted future add beyond auxiliary supervision? **Q3**, is that future physically accurate and causally used? **Q4**, how robust is the method to speed, delay, occlusion, contact, and compute limits? **Q5**, does the benefit transfer to native external protocols, static tasks, and a real conveyor?

A. Setup and Matched Baselines

The proposed matched comparison uses the same demonstrations, frozen DINOv3 features, history mask, action space, optimizer schedule, parameter envelope, controller, and paired evaluation manifest. Its candidate arms are: (1) current-only Action; (2) a history-aware reactive transformer; (3) a validation-selected constant-position, velocity, or acceleration retargeter; (4) *J-Joint*, whose future prediction is auxiliary only; (5) *P-Detach*, the proposed one-way physical interface; (6) *P-E2E*, which permits action gradients into that interface; and (7) future-XYZ and future-$\mathrm{SE}(3)$ oracles. Direct regression and action flow matching would be selected before explicit future information is introduced. External PUMA, DynamicVLA, and PhysMani comparisons require native code and scoring.

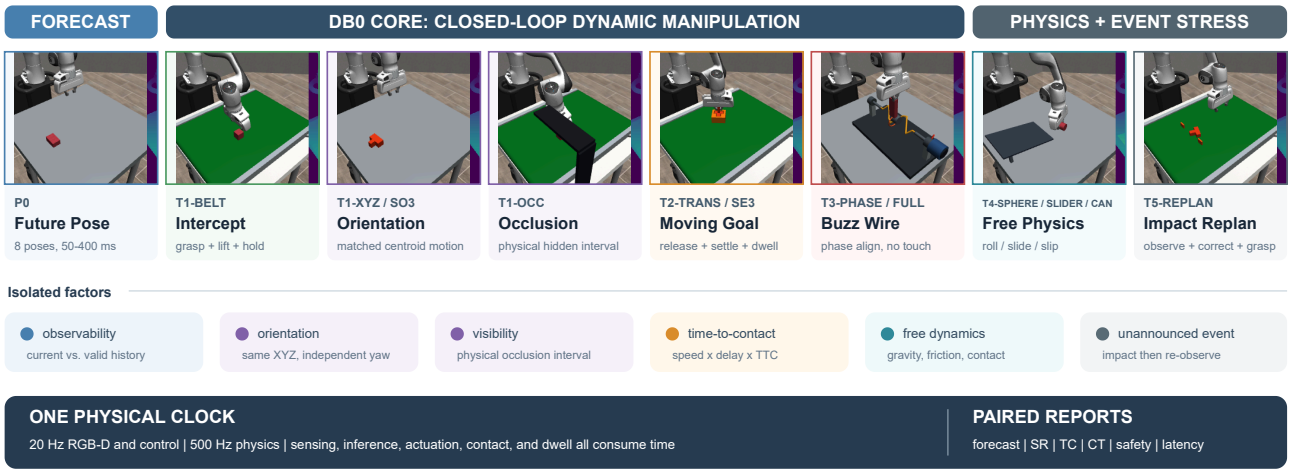


Fig. 3. DB0 tasks and causal factors. P0 measures physical forecasting; T1–T3 form the manipulation Core; T4–T5 stress free dynamics and post-impact replanning. Matched variants isolate orientation, occlusion, and moving-goal geometry while every task shares the same physical clock.

TABLE II

PENDING DB0 COMPARISON ON ONE PAIRED TEST-ID MANIFEST. THE HISTORICAL PLAN PROPOSES FIVE SEEDS AND TASK-BALANCED SUCCESS (SR), TASK COMPLETION (TC), COMPLETION TIME (CT), ORIENTATION/OCCLUSION STRATA, AND UNSAFE CONTACT.

Method	Predicted future	Action reads it?	Core SR (%) ↑	TC ↑	CT (s) ↓	T1-SO3 SR ↑	T1-OCC SR ↑	Unsafe (%) ↓
Current-only Action	None	–	[TBD ± CI]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
History-reactive	None	–	[XX.X]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
Analytic retarget (validation-selected)	XYZ	✓	[TBD ± CI]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
J-Joint	SE(3) auxiliary	–	[TBD ± CI]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
SE(3)-WAM (P-Detach)	SE(3) **+ 3-D flow**	✓	[XX.X]	[X.XXX]	[X.XX]	[TBD]	[TBD]	[X.X]
P-E2E	SE(3) + 3-D flow	✓	[TBD ± CI]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
Future-XYZ oracle	GT XYZ	✓	[TBD ± CI]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
Future-SE(3) oracle	GT SE(3)	✓	[TBD ± CI]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]

TABLE III

PENDING PHYSICAL FORECAST QUALITY AND CAUSAL-USE COMPARISON. THE PLAN USES P0 AND EXOGENOUS PRE-CONTACT WINDOWS; INTERVENTION DELTAS ARE PAIRED WITHIN EXACT-CURRENT GROUPS SO THAT ONLY THE PHYSICAL SUMMARY CHANGES.

Model / intervention	Trans. error (cm) ↓	Sym.-aware rot. (°) ↓	EPE3D (cm) ↓	Flow–SE(3) ↓	Core SR (%) ↑	$\Delta\|\|a\|\|$ ↑	ΔSR (pp)	Interpretation
C1 current-only world	[TBD]	[TBD]	[TBD]	[TBD]	–	–	–	observability floor
C9 direct world	[X.XX]	[X.XX]	[X.XX]	[X.XX]	–	–	–	forecast quality
J-Joint action		same world/data envelope			[TBD]	0 (by design)	0 (by design)	auxiliary supervision
SE(3)-WAM P-Detach		predicted physical summary			[XX.X]	–	–	explicit consumption
swap summary	–	–	–	–	–	[X.XXX]	[-X.X]	exact-current causal test
zero summary	–	–	–	–	–	[TBD]	[-X.X]	information removal
calibrated noise	–	–	–	–	–	[TBD]	[-X.X]	sensitivity
oracle summary	0	0	0	0	[TBD]	[TBD]	[TBD]	controllability ceiling

B. Estimands and Statistical Protocol

Each design claim is paired with an identifying contrast: history length for observability, J-Joint versus P-Detach for forecast consumption, exact-current interventions for causal use, and XYZ versus SE(3) for orientation. The primary estimand is the paired Core-SR difference between P-Detach and the strongest matched history-aware policy. Secondary outcomes are TC, CT conditional on success, unsafe contact, and per-task SR. For an explicit-future method P, history baseline B, and future oracle O, the oracle-gap fraction is

$$G_{\text{oracle}} = \frac{\text{SR}(P) - \text{SR}(B)}{\text{SR}(O) - \text{SR}(B)}. \tag{8}$$

The historical protocol proposes a paired bootstrap and five seed-level values with 100 paired rollouts per seed; these are planned counts, not completed trials. Wilson intervals describe individual SRs; a clustered paired bootstrap resamples exact-current groups within seed for method differences. Horizon, task, visibility, and motion factor are declared strata. Test-OOD, static, external, and real-robot tracks are opened only after model selection and remain separate estimands.

C. Closed-Loop Manipulation

Table II reserves the primary matched comparison. Pending fields are Core SR ([XX.X]%), TC ([X.XXX]), CT ([X.XX] s), matched history-policy SR ([XX.X]%), paired difference

TABLE IV

MATCHED TRAINING AND EVALUATION PROTOCOL.

Item	Contract
DB0 policy packet	2-view RGB-D, 224^2; robot/gripper state; timestamps/calibration are sensor metadata
DB0 record	Step: 2-view RGB-D, robot state, capture alignment, one E7 action; episode: 20/500 Hz, controller, fixed delay, calibration, versions
SE(3)-WAM context	Loader-buffered history/action memory; C1/C2/C3/C9 are matched method ablations
World target	$H = 8$ at 50–400 ms; target $\mathrm{SE}(3)$, patch 3-D displacement, uncertainty
Action	E7, $H_a = 8$; newest-wins continuous execution
Encoder	Frozen DINOv3; depth lift to base-frame XYZ; 4-layer, $d = 256$ MIN tier
Optimization	AdamW, 10^{-4}, cosine decay, 500-step warm-up, effective batch 64
Evidence	3 seeds for screening; 5 for headline; 100 paired rollouts/headline cell

([+X.X] points, 95% CI [X.X, X.X]), and unsafe-contact rate ([X.X]%). The comparison must separately test the value of history, metric 3-D, and P-Detach versus J-Joint under the same world targets and action objective. Per-task and speed–delay–TTC results are needed to rule out gains confined to one easy cell or explained by lower action age.

Future oracles test whether headroom exists beyond a strong history policy. The gap-closed field ([XX]%) remains pending and is interpretable only if the oracle denominator supports a meaningful comparison. Neither P-Detach nor P-E2E is established as superior by this manuscript.

D. Forecast Quality and Causal Use

The pending world-quality fields are [X.XX] cm translation error, [X.XX]° symmetry-aware rotation error, [X.XX] cm 3-D EPE, and [X.XX] Flow–$\mathrm{SE}(3)$ consistency (Table III). Error is reported at each physical horizon and stratified by visibility, speed, acceleration, and rotation. Pose and dense flow have proposed complementary roles: target $\mathrm{SE}(3)$ for rigid grasp alignment, and flow for mover awareness and consistency outside the target region. Those roles need their own controlled evidence.

Exact-current interventions test causal use of the physical bridge. Pending fields record swap action change ([X.XXX]), swap SR change ([-X.X] points), and zero/noise SR changes ([-X.X]/[-X.X] points). A valid intervention holds the current image, robot state, and action-source noise fixed while changing only z. The T1-XYZ/SO3 comparison must test orientation utility with the centroid trajectory held fixed. Action sensitivity alone does not show useful foresight: forecast quality, behavior, safety, and oracle direction must support the same interpretation.

E. Delay Conditions, Systems Profiling, and Transfer

DB0 reports task outcomes at each fixed-delay condition; delay is an ablation axis, not a score. A separate batch-one profile reports [XX.X]/[XX.X] ms p50/p95 loop time, [XX.X] ms p95 action age, and [X.X]% deadline misses; none is read from the offline dataset. Table V reserves separate native DOMINO-SA10, DOM-Dyn4, PhysMani-Dyn16, LIBERO-Static40, and real-conveyor comparisons from the historical plan. No transfer benefit, static compatibility, or real-conveyor success is established by these placeholders.

TABLE V

PENDING TRANSFER AND REAL-TIME EVALUATION. COUNTS ARE HISTORICAL PLANS, NOT COMPLETED TRIALS; EACH NATIVE PROTOCOL REMAINS SEPARATE FROM DB0.

Protocol / metric	N	Reference	SE(3)-WAM	Δ [95% CI]
DOMINO-SA10 SR (%) [3]	1,000	[TBD]	[TBD]	[TBD]
DOM-Dyn4 SR (%) [1]	800	[TBD]	[TBD]	[TBD]
PhysMani-Dyn16 SR (%) [6]	640	[TBD]	[TBD]	[TBD]
LIBERO-Static40 SR (%) [28]	2,000	[TBD]	[TBD]	[TBD]
Real Franka conveyor SR (%)	180	[TBD]	[TBD]	[TBD]
Inference p50/p95 (ms)	–	–	[XX.X]/[XX.X]	–
Action age p95 / miss rate	–	–	[XX.X] ms / [X.X]%	–

Event traces should distinguish reachability failures, occlusion, and responses to unannounced impacts. Their relative frequency and the policy's actual reactions remain unmeasured here; they cannot yet support a failure taxonomy or a claim that action-time alignment improves closed-loop control.

VI. LIMITATIONS AND CONCLUSION

SE(3)-WAM targets single-arm manipulation of task-specified, exogenously moving rigid objects over short horizons. It does not address language-conditioned target selection, long-horizon task planning, deformables, or counterfactual contact outcomes under unseen robot actions. Dense flow after contact is scored only under matched applied actions, and calibrated uncertainty is not itself a safety guarantee. Abrupt unobserved events still require re-observation and replanning.

Within this scope, the proposed experiments test whether dynamic manipulation improves when the policy forecasts the physical world at the time its action will execute. The candidate combines timestamped metric 3-D history, target $\mathrm{SE}(3)$, dense flow, uncertainty, a one-way physical bridge, and newest-wins execution in one feed-forward system. Matched history-aware comparisons, exact-current interventions, and separate external evaluations are needed before claiming that any gain persists or that action generation usefully consumes the forecast. All manuscript results remain pending. The final method and claims must be reconciled with the accepted Research evidence before publication.

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